Assignment #2

ELEC 5404

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Part 1: Training Numerically

The network has one input, two sigmoid hidden neurons, and one linear output. Train using batch-mode backpropagation with $\eta = 0.1$ and no momentum. Deterimine the updated weights after one epoch.

The initial weights are

$$\mathbf{w} = [v_0\ v_1\ v_2\ w_{10}\ w_{11}\ w_{20}\ w_{21}]^T = [0\ \ 4\ \ -1\ \ 0\ \ 2\ \ 1\ \ 0.1]^T.$$

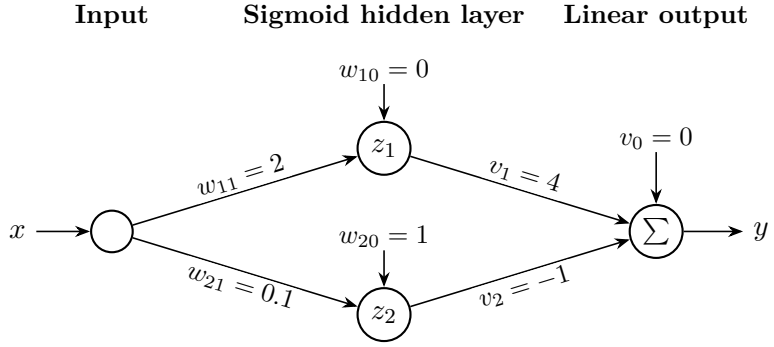


Figure 1: The 1–2–1 MLP with its initial weights.

Table 1: Training data.

i	x_i	d_i
1	0	2
2	1	4

The training error:

$$E(\mathbf{w}) = \frac{1}{2}\sum_{i=1}^{P}\big(y(x_i, \mathbf{w}) - d_i\big)^2.$$

(i) Forward pass of both samples

$$z_1 = \sigma(w_{10} + xw_{11}), \qquad z_2 = \sigma(w_{20} + xw_{21}), \qquad y = v_0 + v_1 z_1 + v_2 z_2.$$

Sample 1 ($x_1 = 0$, $d_1 = 2$):

$$\begin{aligned} z_1 &= \sigma\big(0 + (0)(2)\big) = \sigma(0) = 0.5, \\ z_2 &= \sigma\big(1 + (0)(0.1)\big) = \sigma(1) = 0.7311, \\ y &= 0 + 4(0.5) + (-1)(0.7311) = 1.2689, \\ e_1 &= y - d_1 = 1.2689 - 2 = -0.7311. \end{aligned}$$

Sample 2 ($x_2 = 1$, $d_2 = 4$):

$$\begin{aligned} z_1 &= \sigma\big(0 + (1)(2)\big) = \sigma(2) = 0.8808, \\ z_2 &= \sigma\big(1 + (1)(0.1)\big) = \sigma(1.1) = 0.7503, \\ y &= 0 + 4(0.8808) + (-1)(0.7503) = 2.7729, \\ e_2 &= y - d_2 = 2.7729 - 4 = -1.2271. \end{aligned}$$

(ii) Training error

$$E = \frac{1}{2}\left(e_1^2 + e_2^2\right) = \frac{1}{2}\left((-0.7311)^2 + (-1.2271)^2\right) \quad \Longrightarrow \quad \boxed{E = 1.0201}.$$

(iii) Derivatives w.r.t each weight

Since $E_i = \frac{1}{2}e_i^2$ with $e_i = y_i - d_i$, each per-sample derivative is $\partial E_i/\partial w = e_i\,\partial y_i/\partial w$.

Therefore: $\partial y/\partial w$ for every weight.

Output layer. From $y = v_0 + v_1 z_1 + v_2 z_2$,

$$\frac{\partial y}{\partial v_0} = 1, \qquad \frac{\partial y}{\partial v_1} = z_1, \qquad \frac{\partial y}{\partial v_2} = z_2.$$

Hidden layer. With $\gamma_1 = w_{11}x + w_{10}$, I have $\partial\gamma_1/\partial w_{11} = x$ and $\partial\gamma_1/\partial w_{10} = 1$, and the classic sigmoid derivative being $\sigma'(\gamma) = z(1-z)$. Then applying chain rule,

$$\frac{\partial y}{\partial w_{11}} = \frac{\partial y}{\partial z_1}\frac{\partial z_1}{\partial \gamma_1}\frac{\partial \gamma_1}{\partial w_{11}} = v_1\, z_1(1-z_1)\, x, \qquad \frac{\partial y}{\partial w_{10}} = v_1\, z_1(1-z_1)(1),$$

$$\frac{\partial y}{\partial w_{21}} = \frac{\partial y}{\partial z_2}\frac{\partial z_2}{\partial \gamma_2}\frac{\partial \gamma_2}{\partial w_{21}} = v_2\, z_2(1-z_2)\, x, \qquad \frac{\partial y}{\partial w_{20}} = v_2\, z_2(1-z_2)(1).$$

Multiplying each by e gives the per-sample error derivatives in Table 2.

Table 2: Per-sample error derivatives.

Weight	$\partial E_i/\partial w$
v_0	e
v_1	$e\, z_1$
v_2	$e\, z_2$
w_{10}	$e\, v_1\, z_1(1-z_1)$
w_{11}	$e\, v_1\, z_1(1-z_1)\, x$
w_{20}	$e\, v_2\, z_2(1-z_2)$
w_{21}	$e\, v_2\, z_2(1-z_2)\, x$

Sample 1 ($x_1 = 0$, $e_1 = -0.7311$, $z_1 = 0.5$, $z_2 = 0.7311$):

$$\begin{aligned}
\partial E_1/\partial v_0 &= e_1 = -0.7311,\\
\partial E_1/\partial v_1 &= e_1 z_1 = (-0.7311)(0.5) = -0.3655,\\
\partial E_1/\partial v_2 &= e_1 z_2 = (-0.7311)(0.7311) = -0.5344,\\
\partial E_1/\partial w_{10} &= e_1 v_1 z_1(1-z_1) = (-0.7311)(4)(0.5)(0.5) = -0.7311,\\
\partial E_1/\partial w_{11} &= e_1 v_1 z_1(1-z_1)\, x_1 = (-0.7311)(0) = 0,\\
\partial E_1/\partial w_{20} &= e_1 v_2 z_2(1-z_2) = (-0.7311)(-1)(0.7311)(0.2689) = 0.1437,\\
\partial E_1/\partial w_{21} &= e_1 v_2 z_2(1-z_2)\, x_1 = (0.1437)(0) = 0.
\end{aligned}$$

Sample 2 ($x_2 = 1$, $e_2 = -1.2271$, $z_1 = 0.8808$, $z_2 = 0.7503$):

$$
\begin{aligned}
\partial E_2/\partial v_0 &= e_2 = -1.2271, \\
\partial E_2/\partial v_1 &= e_2 z_1 = (-1.2271)(0.8808) = -1.0808, \\
\partial E_2/\partial v_2 &= e_2 z_2 = (-1.2271)(0.7503) = -0.9206, \\
\partial E_2/\partial w_{10} &= e_2 v_1 z_1 (1 - z_1) = (-1.2271)(4)(0.8808)(0.1192) = -0.5153, \\
\partial E_2/\partial w_{11} &= e_2 v_1 z_1 (1 - z_1)\, x_2 = (-0.5153)(1) = -0.5153, \\
\partial E_2/\partial w_{20} &= e_2 v_2 z_2 (1 - z_2) = (-1.2271)(-1)(0.7503)(0.2497) = 0.2299, \\
\partial E_2/\partial w_{21} &= e_2 v_2 z_2 (1 - z_2)\, x_2 = (0.2299)(1) = 0.2299.
\end{aligned}
$$

(iv) Sum derivatives of both samples and update

$$
w_{\text{new}} = w_{\text{old}} - \eta \, \frac{\partial E}{\partial w}, \qquad \eta = 0.1.
$$

Table 3: Batch gradient and updated weights after one epoch.

Weight	Sample 1	Sample 2	$\partial E/\partial w$	w_{new}
v_0	-0.7311	-1.2271	-1.9581	0.1958
v_1	-0.3655	-1.0808	-1.4463	4.1446
v_2	-0.5344	-0.9206	-1.4551	-0.8545
w_{10}	-0.7311	-0.5153	-1.2464	0.1246
w_{11}	0	-0.5153	-0.5153	2.0515
w_{20}	0.1437	0.2299	0.3737	0.9626
w_{21}	0	0.2299	0.2299	0.0770

The updated weights at the end of the first epoch are therefore

$$
\boxed{\mathbf{w}_{\text{new}} = [v_0\ v_1\ v_2\ w_{10}\ w_{11}\ w_{20}\ w_{21}]^T = [0.1958\ \ 4.1446\ \ -0.8545\ \ 0.1246\ \ 2.0515\ \ 0.9626\ \ 0.0770]^T.}
$$

Part 2: Training by Graphical Illustration

Let the training error and initial weights be

$$E(\mathbf{w}) = w_1^2 - 2w_1 + 1 + 2\left(w_2^2 - 4w_2 + 4\right), \qquad \mathbf{w}(0) = [0.5 \;\; 1]^T.$$

(a) Batch-mode backpropagation with a contour plot

I begin by completing the square in each bracket:

$$E(\mathbf{w}) = \left(w_1^2 - 2w_1 + 1\right) + 2\left(w_2^2 - 4w_2 + 4\right) = (w_1 - 1)^2 + 2(w_2 - 2)^2.$$

Both terms are non-negative, so $E \geq 0$, with $E = 0$ only at the minimum $\mathbf{w}^* = [1 \;\; 2]^T$.

Contours. Setting E equal to a constant c gives

$$c = (w_1 - 1)^2 + 2(w_2 - 2)^2.$$

I then find where this contour crosses the lines through the centre. On the horizontal line $w_2 = 2$,

$$c = (w_1 - 1)^2 \quad \Longrightarrow \quad w_1 = 1 \pm \sqrt{c},$$

and on the vertical line $w_1 = 1$,

$$c = 2(w_2 - 2)^2 \quad \Longrightarrow \quad w_2 = 2 \pm \sqrt{c/2}.$$

Each contour is therefore an ellipse centred at $(1, 2)$, with half-width $\sqrt{c}$ along w_1 and half-height $\sqrt{c/2}$ along w_2.

Initial point. At $\mathbf{w}(0)$,

$$E(\mathbf{w}(0)) = (0.5 - 1)^2 + 2(1 - 2)^2 = 0.25 + 2 = 2.25,$$

so the $E = 2.25$ contour passes through $\mathbf{w}(0)$. Its points follow from choosing w_1 and solving $w_2 = 2 \pm \sqrt{(2.25 - (w_1 - 1)^2)/2}$:

Table 4: Points on the $E = 2.25$ contour.

w_1	-0.5	0	0.5	1	1.5	2	2.5
w_2 (top)	2	2.79	3	3.06	3	2.79	2
w_2 (bottom)	2	1.21	1	0.94	1	1.21	2

Epoch 1

Gradient. Differentiating the expanded error $E = w_1^2 - 2w_1 + 1 + 2w_2^2 - 8w_2 + 8$ with respect to each weight gives

$$\frac{\partial E}{\partial w_1} = 2w_1 - 2, \qquad \frac{\partial E}{\partial w_2} = 4w_2 - 8.$$

At $\mathbf{w}(0) = [0.5 \;\; 1]^T$,

$$\frac{\partial E}{\partial w_1} = 2(0.5) - 2 = -1, \qquad \frac{\partial E}{\partial w_2} = 4(1) - 8 = -4 \quad \Longrightarrow \quad \nabla E = [-1 \;\; -4]^T.$$

Direction. With zero momentum, the backpropagation direction is the negative gradient:

$$\mathbf{h} = -\nabla E = [1 \;\; 4]^T.$$

Line minimization. Starting at $\mathbf{w}(0)$, each point along $\mathbf{h}$ is

$$\mathbf{w} = \mathbf{w}(0) + \eta\,\mathbf{h} = [0.5 \;\; 1]^T + \eta\,[1 \;\; 4]^T \quad \Longrightarrow \quad w_1 = 0.5 + \eta, \qquad w_2 = 1 + 4\eta.$$

Evaluating E at trial step sizes shows that it decreases and then rises again, with the minimum near $\eta \approx 0.25$:

Table 5: Trial step sizes along $\mathbf{h}$ in epoch 1.

η	w_1	w_2	E
0	0.5	1	2.25
0.1	0.6	1.4	0.88
0.2	0.7	1.8	0.17
0.25	**0.75**	**2**	**0.0625**
0.3	0.8	2.2	0.12
0.4	0.9	2.6	0.73
0.5	1	3	2

To find the exact minimum, I substitute the line into E:

$$E(\eta) = (0.5 + \eta - 1)^2 + 2(4\eta - 1)^2 = 33\eta^2 - 17\eta + 2.25.$$

Setting the derivative to zero,

$$\frac{dE}{d\eta} = 66\eta - 17 = 0 \quad \Longrightarrow \quad \eta = \frac{17}{66} = 0.2576.$$

The new weights at the end of epoch 1 are therefore

$$w_1 = 0.5 + 0.2576 = 0.7576,$$
$$w_2 = 1 + 4(0.2576) = 2.0303,$$

$$\boxed{\mathbf{w}(1) = [0.7576 \;\; 2.0303]^T}, \qquad E = (0.7576 - 1)^2 + 2(2.0303 - 2)^2 = 0.0606.$$

Epoch 2

Gradient and direction. At $\mathbf{w}(1)$,

$$\frac{\partial E}{\partial w_1} = 2(0.7576) - 2 = -0.4848, \qquad \frac{\partial E}{\partial w_2} = 4(2.0303) - 8 = 0.1212,$$

$$\nabla E = [-0.4848 \;\; 0.1212]^T, \qquad \mathbf{h} = -\nabla E = [0.4848 \;\; -0.1212]^T.$$

Line minimization. Each point along the new direction is

$$\mathbf{w} = [0.7576 \;\; 2.0303]^T + \eta\,[0.4848 \;\; -0.1212]^T \quad \Longrightarrow \quad w_1 = 0.7576 + 0.4848\eta, \qquad w_2 = 2.0303 - 0.1212\eta.$$

Table 6: Trial step sizes along $\mathbf{h}$ in epoch 2.

η	w_1	w_2	E
0	0.7576	2.0303	0.0606
0.2	0.8546	2.0061	0.0212
0.4	0.9515	1.9818	0.0030
0.5	**1**	**1.9697**	**0.0018**
0.6	1.0485	1.9576	0.0059
0.8	1.1454	1.9333	0.0300

The minimum lies between $\eta = 0.4$ and $\eta = 0.5$. Substituting the line into E,

$$E(\eta) = (0.4848\eta - 0.2424)^2 + 2(0.0303 - 0.1212\eta)^2 = 0.2644\eta^2 - 0.2497\eta + 0.0606,$$

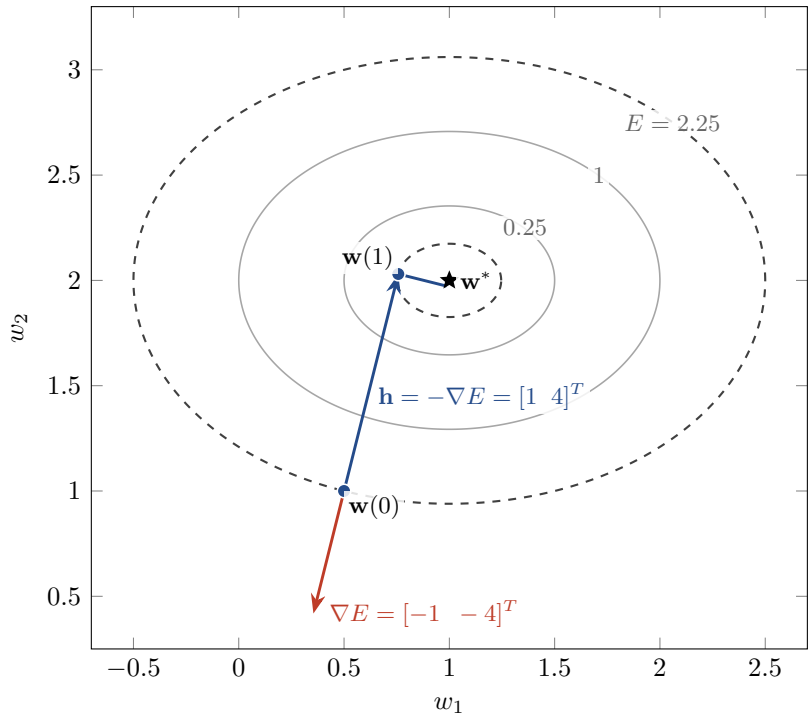
$$\frac{dE}{d\eta} = 0.5288\eta - 0.2497 = 0 \quad \Longrightarrow \quad \eta = \frac{0.2497}{0.5288} = 0.4722.$$

The weights at the end of epoch 2 are therefore

$$\boxed{\mathbf{w}(2) = [0.9865 \;\; 1.9731]^T}, \qquad E = 0.00163.$$

Contour plot. Figure 2 shows the contours, the initial point, the gradient and search direction in each epoch, and the location of $\mathbf{w}$ after each epoch. Each line search ends where the search line is tangent to a contour. The two search directions are perpendicular, since $[1 \;\; 4] \cdot [0.4848 \;\; -0.1212] = 0$. The path therefore zig-zags, and after two epochs $\mathbf{w}(2)$ is close to, but not at, the minimum $\mathbf{w}^* = [1 \;\; 2]^T$.

(a) Epoch 1



(b) Epoch 2 (zoom near $\mathbf{w}^*$)

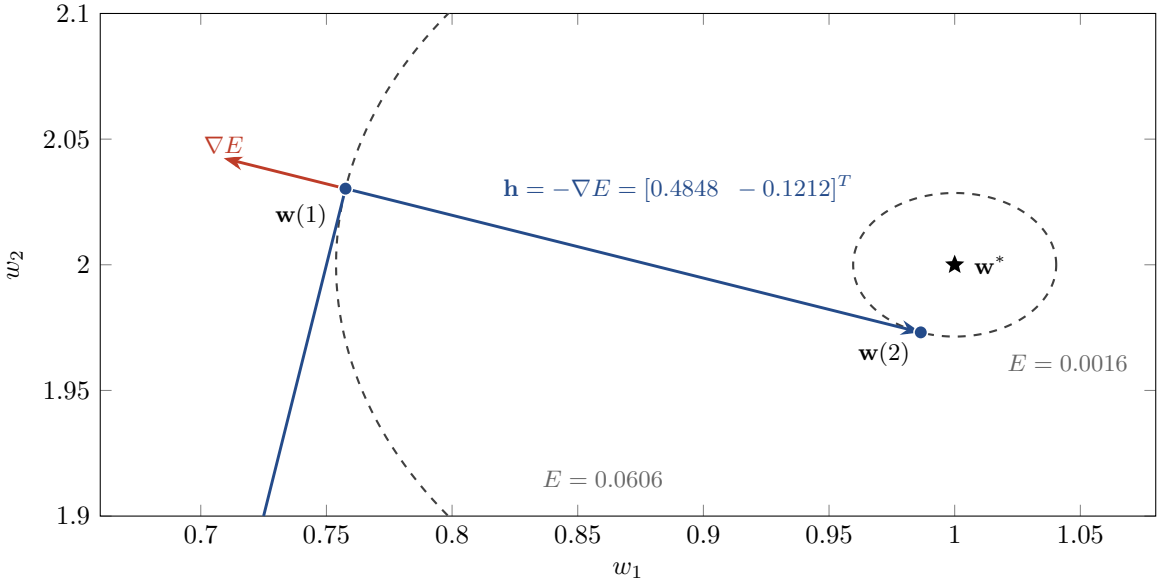


Figure 2: Batch-mode backpropagation with line minimization. Dashed contours pass through $\mathbf{w}(0)$, $\mathbf{w}(1)$ and $\mathbf{w}(2)$. Red arrows show ∇E and blue arrows show $\mathbf{h} = -\nabla E$ (directions only, not to scale). Panel (b) enlarges the region around $\mathbf{w}^*$ to show epoch 2.

(b) Conjugate gradient method

1. Number of epochs. Two epochs are needed, because there are two weights. E is a quadratic function of the $n = 2$ weights, and with exact line minimization, the conjugate gradient method reaches

the minimum of an n-weight quadratic in at most n epochs. This is confirmed below, where $E = 0$ at the end of epoch 2.

2. Epoch 1. With no previous direction, the first conjugate gradient epoch is identical to epoch 1 of part (a):

$$\nabla E(\mathbf{w}(0)) = [-1 \quad -4]^T, \qquad \mathbf{h}_0 = -\nabla E = [1 \;\; 4]^T, \qquad \eta = 0.2576,$$
$$\mathbf{w}(1) = [0.7576 \;\; 2.0303]^T, \qquad E = 0.0606.$$

3. Negative gradient in epoch 2. Epoch 2 begins as in part (a):

$$\frac{\partial E}{\partial w_1} = 2(0.7576) - 2 = -0.4848, \qquad \frac{\partial E}{\partial w_2} = 4(2.0303) - 8 = 0.1212,$$
$$\nabla E(\mathbf{w}(1)) = [-0.4848 \;\; 0.1212]^T \quad \Longrightarrow \quad -\nabla E(\mathbf{w}(1)) = [0.4848 \quad -0.1212]^T.$$

4. Conjugate gradient direction. The new direction combines the negative gradient with the previous direction:

$$\mathbf{h}_1 = -\nabla E(\mathbf{w}(1)) + \gamma\, \mathbf{h}_0.$$

Using the Fletcher–Reeves formula,

$$\gamma = \frac{|\nabla E(\mathbf{w}(1))|^2}{|\nabla E(\mathbf{w}(0))|^2} = \frac{(-0.4848)^2 + (0.1212)^2}{(-1)^2 + (-4)^2} = \frac{0.2497}{17} = 0.0147.$$

Therefore,

$$\mathbf{h}_1 = [0.4848 \;\; -0.1212]^T + 0.0147\,[1 \;\; 4]^T = [0.4848 \;\; -0.1212]^T + [0.0147 \;\; 0.0588]^T = [0.4995 \;\; -0.0624]^T.$$

5. Line minimization and the location of w after epoch 2. Each point along $\mathbf{h}_1$ is

$$\mathbf{w} = [0.7576 \;\; 2.0303]^T + \eta\,[0.4995 \;\; -0.0624]^T \quad \Longrightarrow \quad w_1 = 0.7576 + 0.4995\eta, \qquad w_2 = 2.0303 - 0.0624\eta.$$

Table 7: Trial step sizes along the conjugate direction $\mathbf{h}_1$ in epoch 2.

η	w_1	w_2	E
0	0.7576	2.0303	0.0606
0.2	0.8575	2.0178	0.0209
0.4	0.9574	2.0053	0.0019
0.5	**1.0073**	**1.9991**	**0.0001**
0.6	1.0573	1.9928	0.0034
0.8	1.1572	1.9803	0.0255

The minimum lies near $\eta = 0.5$, where E is already almost zero. Substituting the line into E,

$$E(\eta) = (0.4995\eta - 0.2424)^2 + 2(0.0303 - 0.0624\eta)^2 = 0.25729\eta^2 - 0.24972\eta + 0.06061,$$
$$\frac{dE}{d\eta} = 0.51458\eta - 0.24972 = 0 \quad \Longrightarrow \quad \eta = \frac{0.24972}{0.51458} = 0.4853.$$

The weights at the end of epoch 2 are therefore

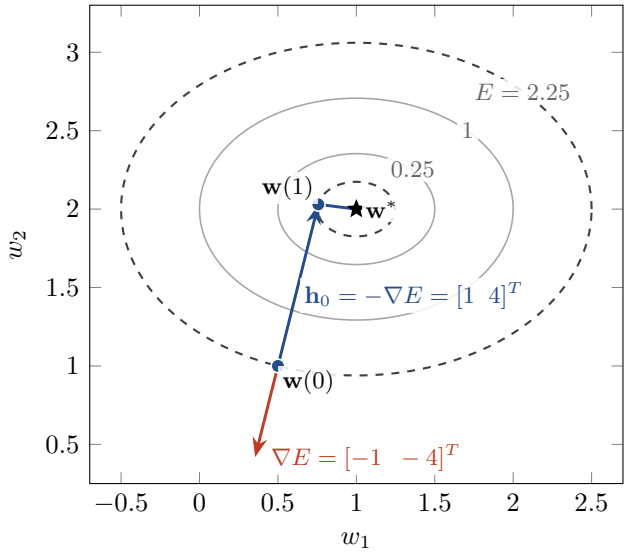
$$w_1 = 0.7576 + 0.4995(0.4853) = 0.7576 + 0.2424 = 1.0000,$$
$$w_2 = 2.0303 - 0.0624(0.4853) = 2.0303 - 0.0303 = 2.0000,$$

$$\boxed{\mathbf{w}(2) = [1 \;\; 2]^T = \mathbf{w}^*}, \qquad E = (1-1)^2 + 2(2-2)^2 = 0.$$

The training error reaches zero after two epochs, as stated in item 1.

Contour plot. Figure 3 shows the conjugate gradient training. Epoch 1 is the same as in Figure 2. In epoch 2, adding $\gamma \mathbf{h}_0$ to $-\nabla E$ tilts the direction from $[4 \;\; -1]$ to $[0.4995 \;\; -0.0624] \propto [8 \;\; -1]$. This is exactly the direction from $\mathbf{w}(1)$ to the minimum, $\mathbf{w}^* - \mathbf{w}(1) = [0.2424 \;\; -0.0303]^T$. The two search directions are conjugate rather than perpendicular: $\mathbf{h}_0^T \operatorname{diag}(2,4)\, \mathbf{h}_1 = 2(1)(0.4995) + 4(4)(-0.0624) \approx 0$.

(a) Epoch 1



(b) Epoch 2 (zoom near $\mathbf{w}^*$)

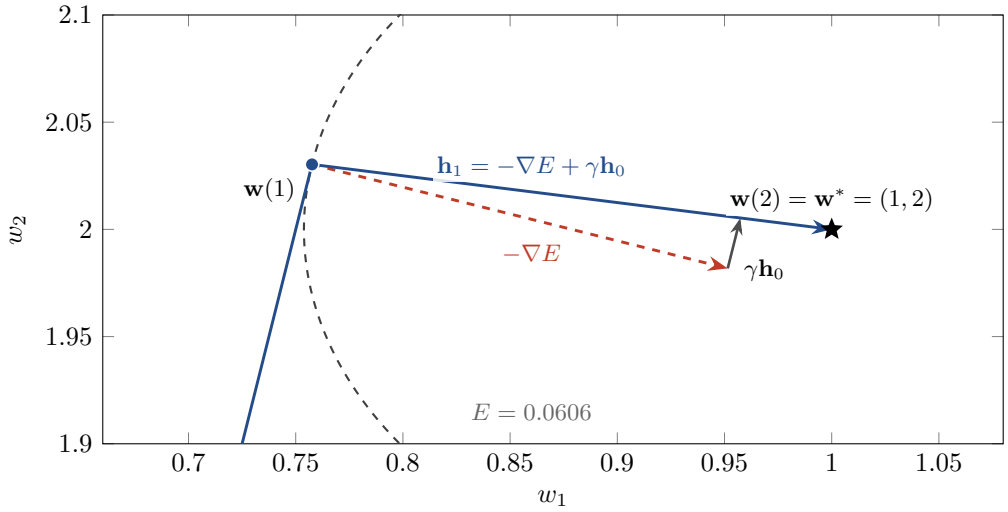


Figure 3: Conjugate gradient training with line minimization. Epoch 1 is identical to part (a). In epoch 2, the dashed red arrow is the negative gradient, the grey arrow is $\gamma \mathbf{h}_0$ (vectors scaled for visibility), and their sum $\mathbf{h}_1$ leads directly to the minimum, so $E = 0$ after two epochs.